
AI-Powered DDoS Attack Detection

Using Temporal Deep Learning for Network Security

Paper Draft — Executive Summary

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March 2026

Executive Summary

DDoS attacks cost enterprises an average of \$218,000 per incident and are growing in sophistication. I built an AI-powered intrusion detection system that classifies network traffic as Attack, Benign, or Suspicious with 94% accuracy—recovering an entire threat category that conventional tools miss completely. The system reads sequences of network traffic rather than isolated snapshots, detecting temporal attack patterns invisible to traditional approaches. A confidence-based escalation mechanism auto-classifies 85% of traffic at 96%+ accuracy while routing ambiguous cases for human review.

Recommendation: Deploy the ensemble model with the 0.75 confidence threshold—reducing analyst workload by 85% while maintaining near-zero missed attack rate and shifting from reactive incident response to proactive threat prevention.

Key Metric	Achieved	Significance
Detection Accuracy	94.1%	Exceeds 80% target by 14 pts
Suspicious Traffic Recall	93.6%	Catches 9 of 10 ambiguous threats
Auto-Classification Coverage	85%	Remaining cases go to analysts
High-Confidence Accuracy	96%+	On the auto-classified portion
Data Integrity	Zero leakage	Results reflect real-world conditions

1. The Problem

DDoS attacks overwhelm network infrastructure with malicious traffic, causing six-figure average incident costs in downtime, reputational damage, and recovery. Conventional security tools examine each connection as an independent event and cannot detect multi-phase attack sequences—they either miss novel threats or drown security teams in spurious alerts. Industry leader Cloudflare addresses this at scale using temporal baselines built from seven-day rolling windows, achieving 98.6% detection (Cloudflare, 2024). This project applies the same temporal principle at the flow level for organizations without Cloudflare-scale infrastructure.

2. Background

Current detection approaches span a wide accuracy-versus-cost spectrum. Transformer-based DDoSBERT (Le et al., 2025) achieves top-tier accuracy on the BCCC dataset but demands computational resources that rule out edge deployment. LUCID (Doriguzzi-Corin et al., 2020) processes over 55,000 samples per second with only 2,241 parameters but classifies each window independently—it cannot track attack progression. A comparative study by Wolosewicz et al. (2024) on the same dataset revealed that a standard neural network posted strong headline accuracy yet failed on the majority of held-out attack samples by exploiting class distribution. This project targets the gap between these extremes: competitive detection on standard hardware with temporal awareness across consecutive windows.

3. The Approach

I developed an AI detection pipeline that groups network connections into time-ordered sequences and trains two complementary models to classify traffic into three categories:

Category	What It Means
Attack	Confirmed malicious traffic requiring immediate automated blocking
Benign	Normal, legitimate traffic—no action required
Suspicious	Ambiguous traffic (possible early-stage attack activity) flagged for analyst review

Most existing tools force a binary Attack/Benign decision. The three-class design is intentional: the Suspicious category surfaces ambiguous traffic that may signal the opening moves of an attack, giving security teams advance notice before an incident materializes.

3.1 How It Works (Non-Technical Overview)

1. Data preparation: 700,774 network connections from the BCCC-Cloud-DDoS-2024 dataset (University of New Brunswick, 2024) are cleaned, time-sorted, and distilled from 318 raw measurements to the 86 most informative indicators.
2. Temporal grouping: Connections are assembled into sliding windows of 30 consecutive events, creating sequences that reveal how traffic behavior shifts over time.
3. Dual-model detection: Two AI models analyze each window from different angles—one tuned for long-range sequential dependencies, the other for localized burst signatures.
4. Ensemble decision: Both models’ probability outputs are merged into a single classification with an associated confidence score.
5. Escalation routing: Clear-cut decisions execute automatically; uncertain cases route to a human analyst.

4. Requirements Specification

4.1 Functional Requirements

Functional requirements define what the system must do—the capabilities it must deliver to be considered complete.

ID	Requirement	How It Is Met	Status
FR-1	Ingest BCCC-Cloud-DDoS-2024 flows and preserve temporal ordering	700,774 flows sorted chronologically by timestamp before any processing	Met
FR-2	Convert raw flows into temporal sequences that capture attack progression	30-flow sliding windows, stride 15, last-flow labeling	Met
FR-3	Classify each window into Attack, Benign, or Suspicious with a confidence score	Soft-vote ensemble of BiLSTM + 1D CNN; softmax probabilities per class	Met
FR-4	Handle class imbalance so the model learns attack patterns, not data distribution	Class-weighted loss, label smoothing (0.1), stratified splits	Met
FR-5	Route low-confidence classifications to a human analyst for review	Confidence threshold at 0.75; windows below threshold enter analyst queue	Met

4.2 Non-Functional Requirements

Non-functional requirements define how well the system must perform—quality attributes covering performance, security, and operational constraints.

Performance

ID	Requirement	Target / Achieved	Status
P-1	Overall 3-class test accuracy	≥ 80% / Achieved 94.11%	Exceeded
P-2	Attack and Benign F1-score	≥ 0.94 / Achieved ≥ 0.94 both classes	Met
P-3	Suspicious recall (catch early-warning threats)	≥ 85% / Achieved 93.6%	Exceeded
P-4	Computational cost lower than transformer-scale models	481K–688K params vs. DDoSBERT; single GPU trainable	Met

Security

ID	Requirement	Implementation	Status
S-1	Zero data leakage between training and test splits	Scaler and feature selectors fitted on training-time rows only	Met
S-2	Errors must favor over-alerting, not missed attacks	Confusion matrix confirms Suspicious→Attack as dominant error	Met

Constraints

ID	Constraint	Detail
C-1	Single dataset	Trained and tested on BCCC-Cloud-DDoS-2024 only; generalization to other networks unverified
C-2	Hardware ceiling	All training performed on a single NVIDIA Quadro P5000 GPU (16 GB VRAM)
C-3	Fixed temporal window	30-flow window; may miss attacks shorter or longer than this span
C-4	Offline evaluation only	Evaluated on historical data; real-time streaming inference not yet implemented
C-5	Class imbalance	Suspicious is 8.4% of flows; precision capped at ~70% despite mitigation strategies
C-6	Open-source stack only	Python 3, PyTorch, scikit-learn, NumPy, Pandas; no proprietary dependencies

5. Preliminary Results

5.1 Detection Performance

Model	Accuracy	Attack F1	Benign F1	Suspicious F1
Sequential Model (BiLSTM)	93.84%	0.945	0.959	0.795
Pattern Model (1D CNN)	94.11%	0.943	0.963	0.805
Combined Ensemble	≥ 94.11%	≥ 0.945	≥ 0.963	Improved
Previous Baseline	~47%	—	—	0% detected

F1 combines precision and recall into a single score; 1.0 is perfect.

Both models exceed the 80% target by a wide margin. The temporal approach succeeds where the standard neural network tested by Wolosewicz et al. (2024) failed: by training on traffic sequences with explicit class balancing, the system detects the minority Suspicious class that headline-accuracy models routinely miss.

5.2 Escalation and Strategic Value

Inspired by Cloudflare's tiered architecture (Cloudflare, 2024), the system routes only low-confidence verdicts to analysts while automating clear-cut decisions. This addresses the ~70% Suspicious precision limitation by ensuring human review is reserved for genuinely ambiguous cases. The Suspicious category itself provides an early-warning signal that binary systems cannot offer, and every verdict carries a confidence score for audit and compliance purposes.

6. Validation

Seven validation scenarios passed: holdout accuracy (94.11%), per-category F1 (≥ 0.94 for Attack/Benign), misclassification direction (favors caution over missed threats), overfitting check (train/val convergence), ensemble comparison (matches or exceeds both models), escalation viability (coverage/accuracy trade-off confirmed), and data pipeline integrity (zero leakage). When errors occur, the system errs toward caution—flagging benign traffic rather than letting attacks pass undetected.

7. What Comes Next

1. Final paper: Complete expanded literature review, deployment architecture, and conclusion in Weeks 11–12.

References

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